香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

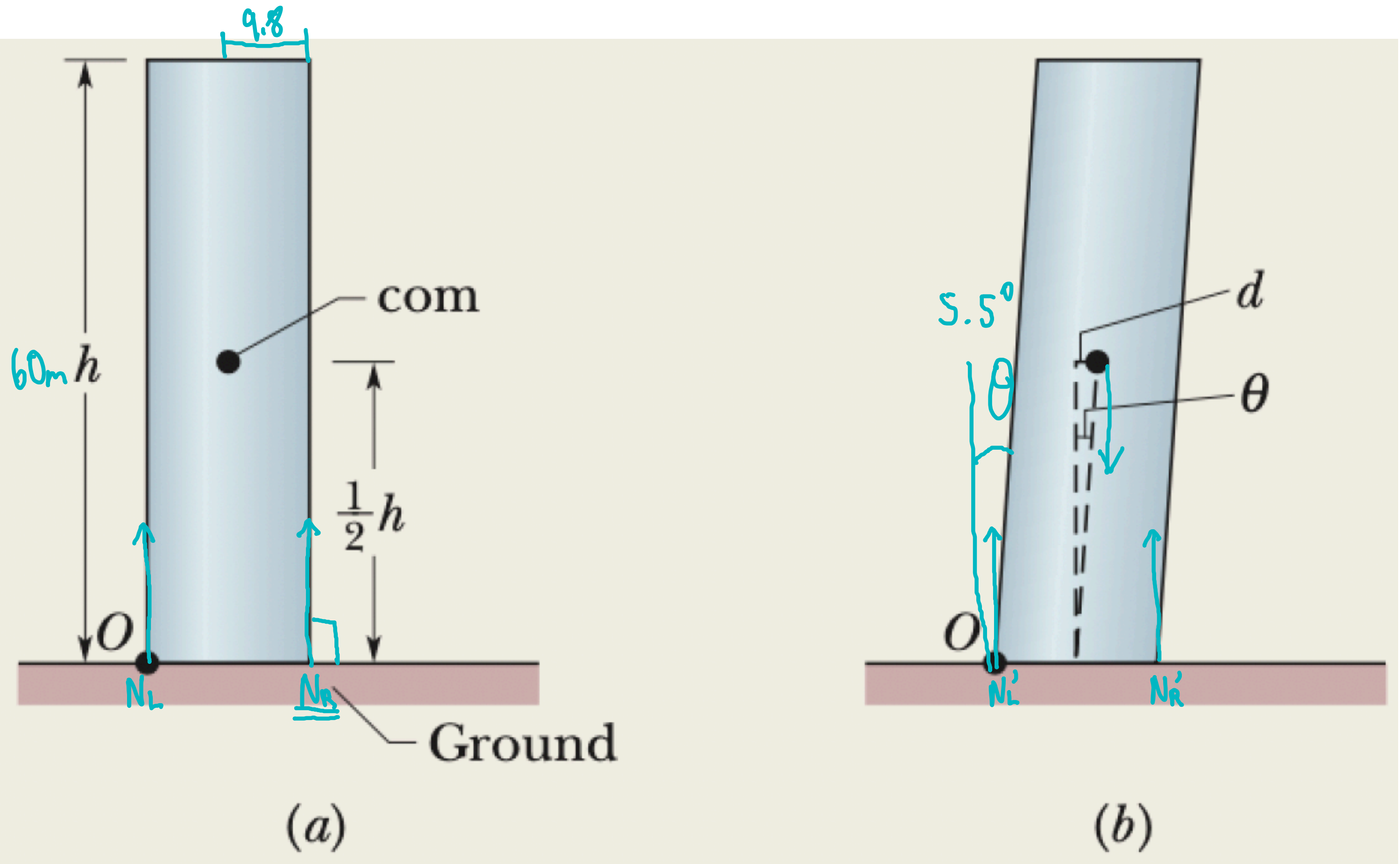
Lecture 16: Equilibrium & Elasticity

Jingqiang Ye
School of Science and Engineering (SSE)
March 20th, 2025

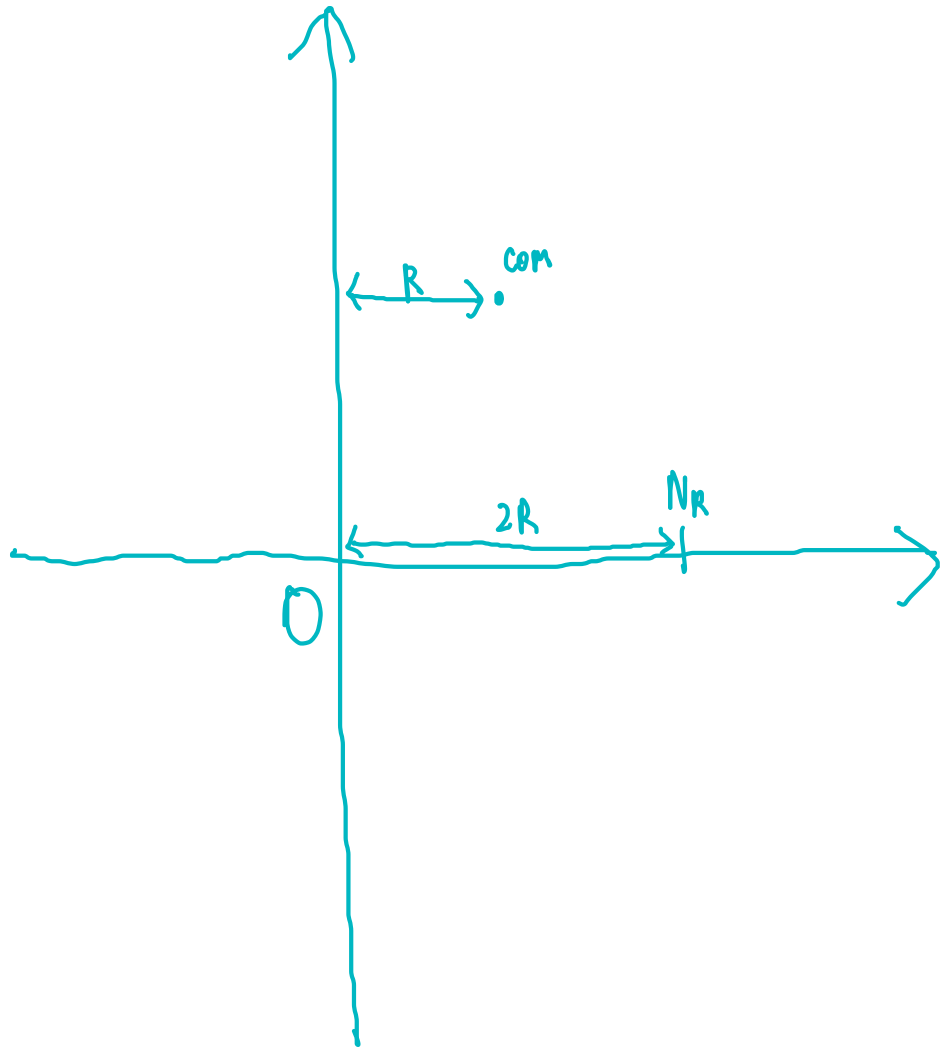
Recap: Equilibrium

- Condition of equilibrium:
  - Net force is zero, i.e. $\sum_i \vec{F}_i = 0$
  - Net torque is zero, i.e. $\sum_i \vec{\tau}_i = 0$
- Two types of equilibrium
  - Stable equilibrium: minimum in energy
  - Unstable equilibrium: maximum in energy

Balancing the Leaning Tower of Pisa



Let's assume that the Tower of Pisa is a uniform hollow cylinder of radius R = 9.8 m and height h = 60 m. The center of mass is located at height h/2, along the cylinder's central axis. In Fig.a, the cylinder is upright. In Fig. b, it leans rightward (toward the tower's southern wall) by $\theta = 5.5^o$, which shifts the com by a distance d. Let's assume that the ground exerts only two forces on the tower. A normal force N_L acts on the left (northern) wall, and a normal force N_R acts on the right (southern) wall. By what percent does the magnitude N_R increase because of the leaning?



∵ building no move → eq

$\sum \tau_i = 0$

$mgR = N_R \cdot 2R$

$N_R = \frac{mg}{2}$

$N_R' \cdot 2R = mg(R+d)$

$d = \frac{h}{2}\tan\theta$

$N_R' \cdot 2R = mg(R + \frac{h}{2}\theta)$

Small angle approx.
$\sin\theta = \tan\theta = \theta = \frac{5.5^o}{360^o} \cdot 2\pi$
θ in radian

$N_R' = \frac{mg}{2}(1 + \frac{h\theta}{2R})$

$\frac{N_R'}{N_R} - 1 = \frac{h\theta}{2R} \approx 21\%$

Hooke's Law

$$F = -kx$$

Force

Spring constant

Deformation

Non-rigid Body

- Rigid body cannot deform, but real body can!

Stress = Modulus × Strain

	Definition	Unit
Stress	Force/area	$\mathrm{N/m^2} = \mathrm{Pa}$ (Pressure)
Strain	Fractional deformation	Dimensionless
Elastic modulus	Stress/strain	$\mathrm{N/m^2} = \mathrm{Pa}$ (Pressure)

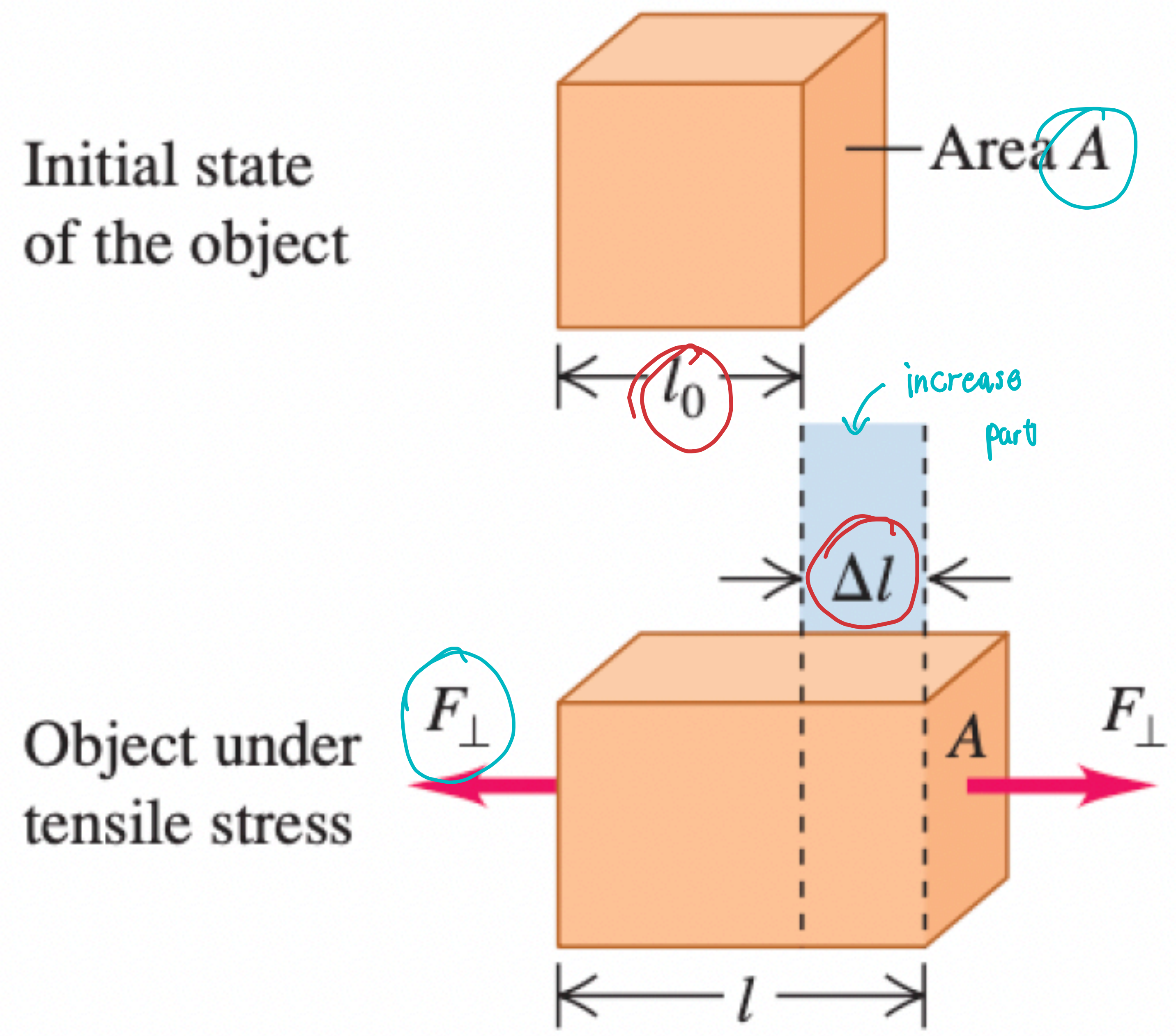
Non-rigid Body

- Stretch/compress
- Twist
- Squeeze

Young's Modulus

- Compress/Stretch

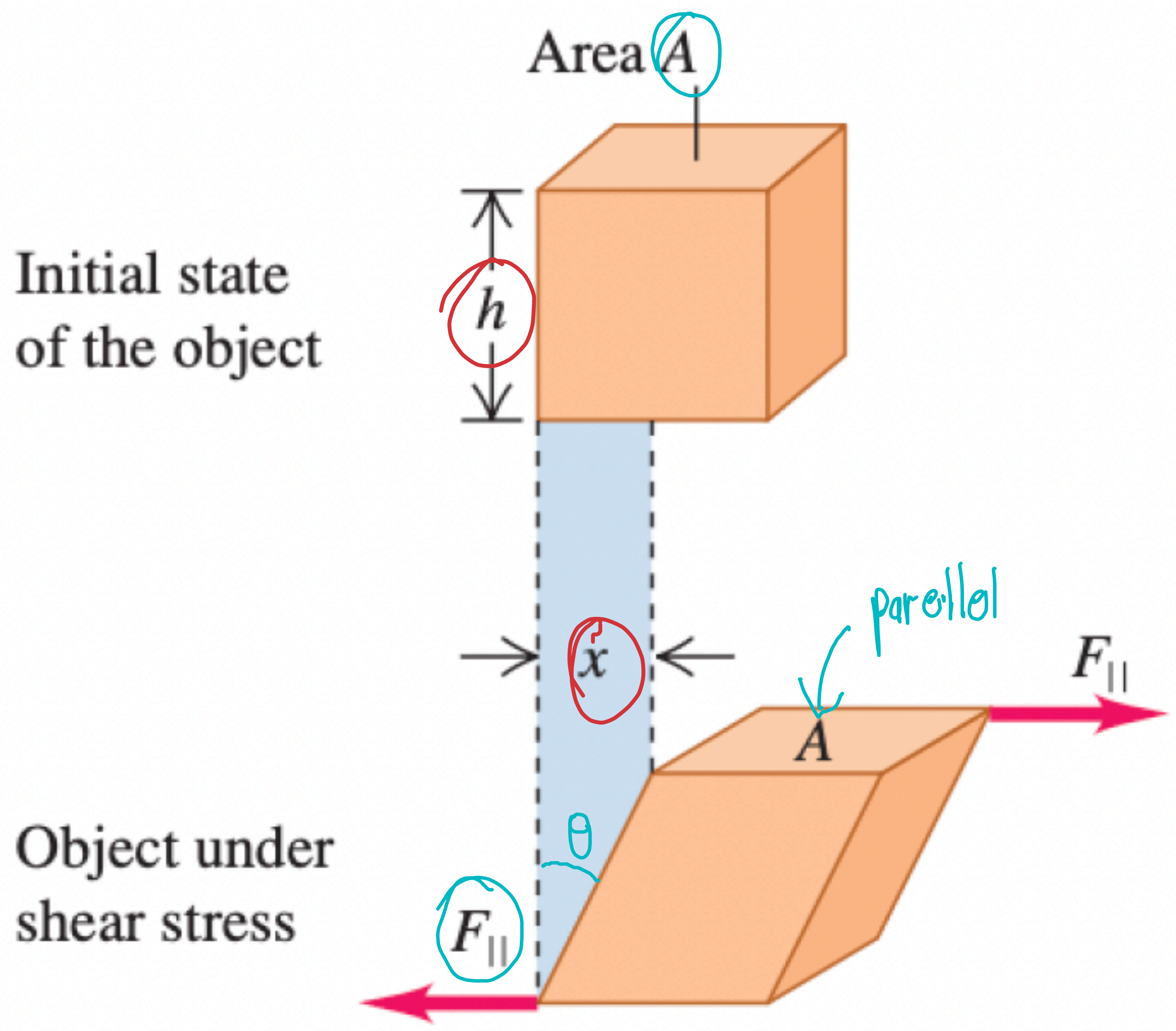
	Definition
Tensile Stress	$\dfrac{F_\perp}{A}$
Tensile Strain	$\dfrac{\Delta l}{l_0}$
Young's modulus	$\dfrac{F_\perp/A}{\Delta l/l_0} = \dfrac{F_\perp l_0}{A\Delta l} = Y$



Shear Modulus

- Twist

	Definition
Shear Stress	$\frac{F_{\parallel}}{A}$
Shear Strain	$\frac{x}{h}\,(x \ll h)$
Shear modulus	$\frac{F_{\parallel}/A}{x/h} = \frac{F_{\parallel}h}{Ax} = S$



$\frac{x}{h} = \tan\theta = \theta$

Bulk Modulus

Definition

increase pressure

Stress $\Delta P = \frac{F}{A}$

Strain $\frac{\Delta V}{V}$

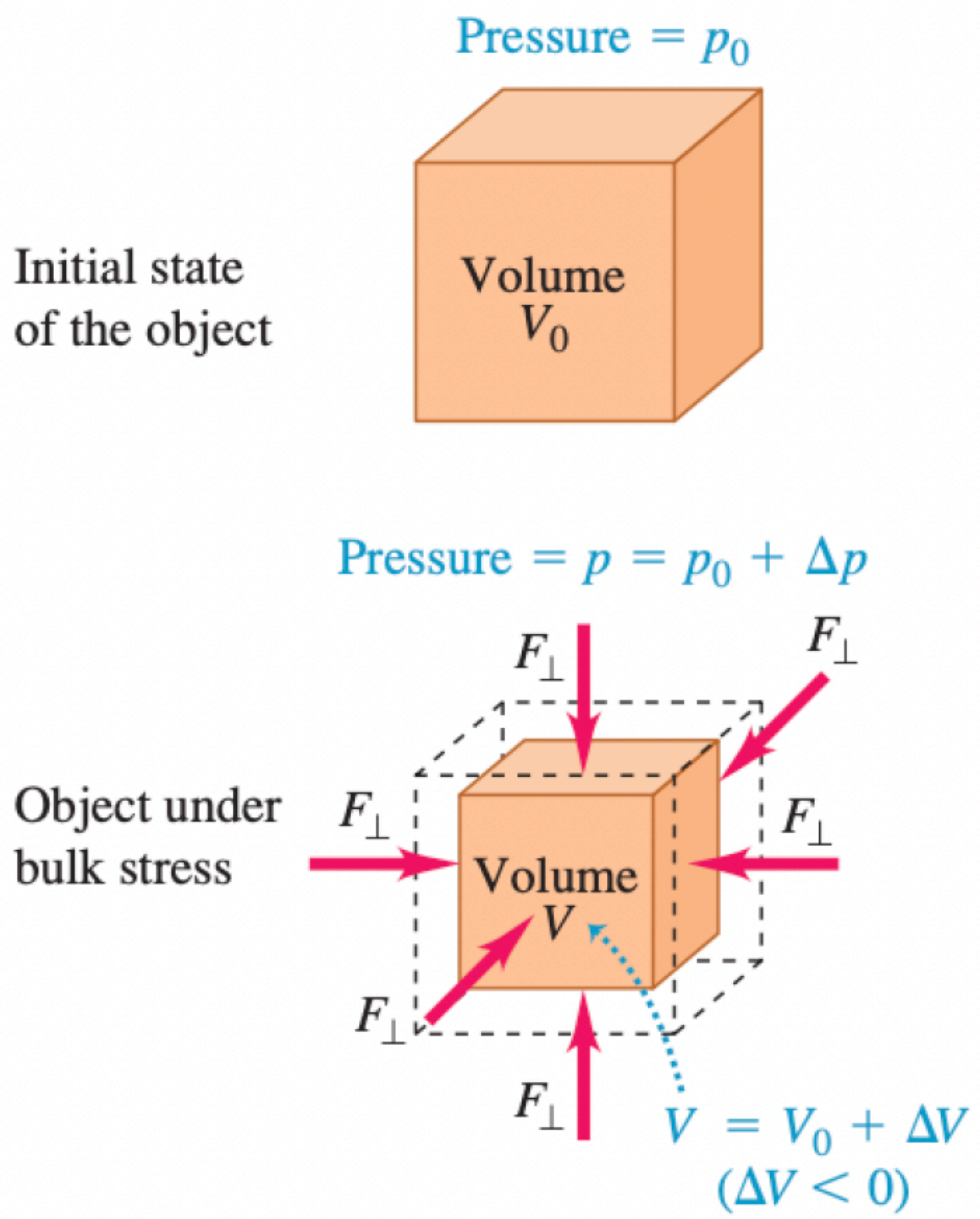
Bulk modulus $B = -\frac{\Delta P}{\Delta V / V}$

↳ make sure positive

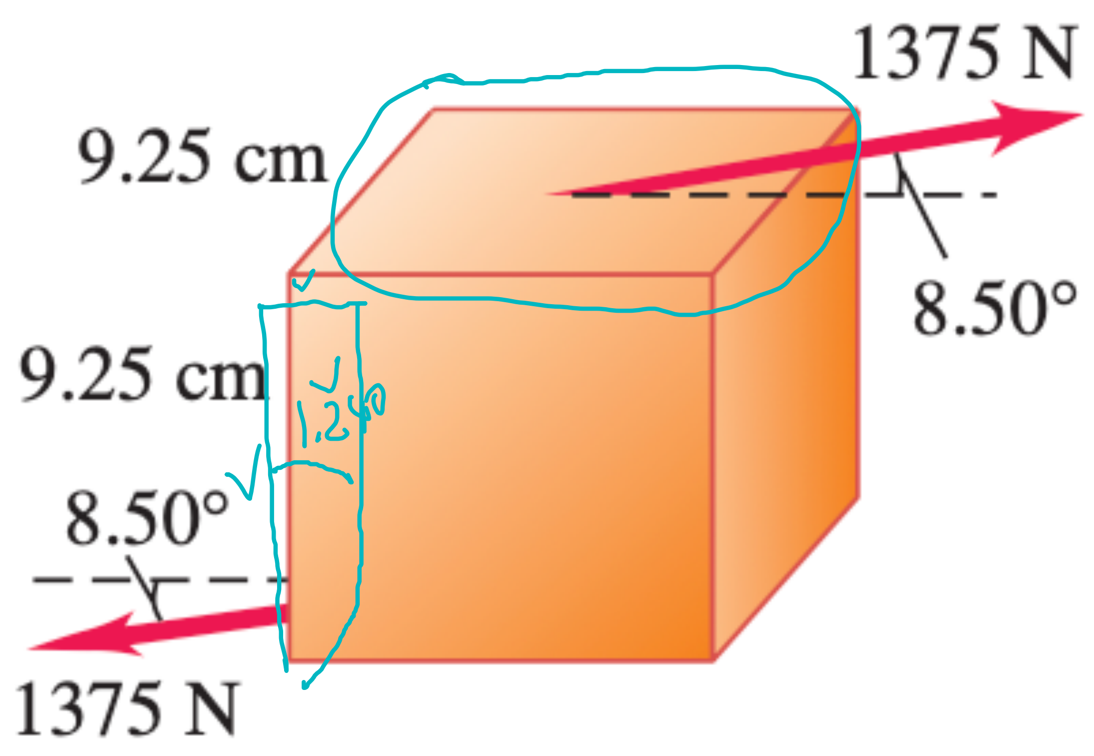
+

−

Compressibility $k = \frac{1}{B}$



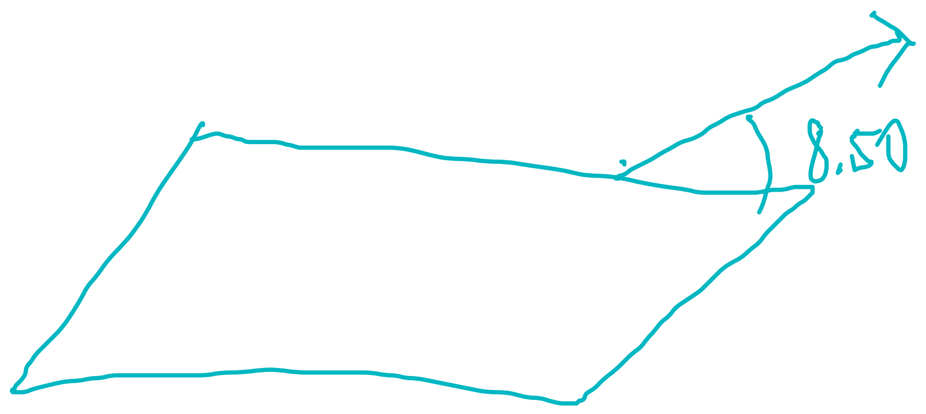
Exercise



In lab tests on a 9.25-cm cube of a certain material, a force of 1375 N directed at 8.50° to the cube causes the cube to deform through an angle of 1.24°, as shown in the left figure. What is the shear modulus of the material?

$$\text{Stress} = \frac{1375 \cos 8.50^\circ}{(9.25 \div 100)^2}$$

$$\text{Strain} = \frac{9.25 \times \frac{1.24^\circ}{360^\circ} \times 2\pi}{9.25 \div 100}$$



$$S = \frac{\text{Stress}}{\text{Strain}} = 7.344 \times 10^6 \text{ Pa}$$

$$= 734.4 \text{ N/cm}^2$$

Hydraulic Press

A hydraulic press contains 0.25 m^3 of oil. The bulk modulus of the oil is $B = 5.0 \times 10^9$ Pa. When the oil is subject to a pressure increase $\Delta P = 1.6 \times 10^7$ Pa, find the decrease in the volume of the oil.

$$B = -\frac{\Delta P}{\Delta V / V}$$

$$5.0 \times 10^9 = \frac{-1.6 \times 10^7}{\Delta V / V}$$

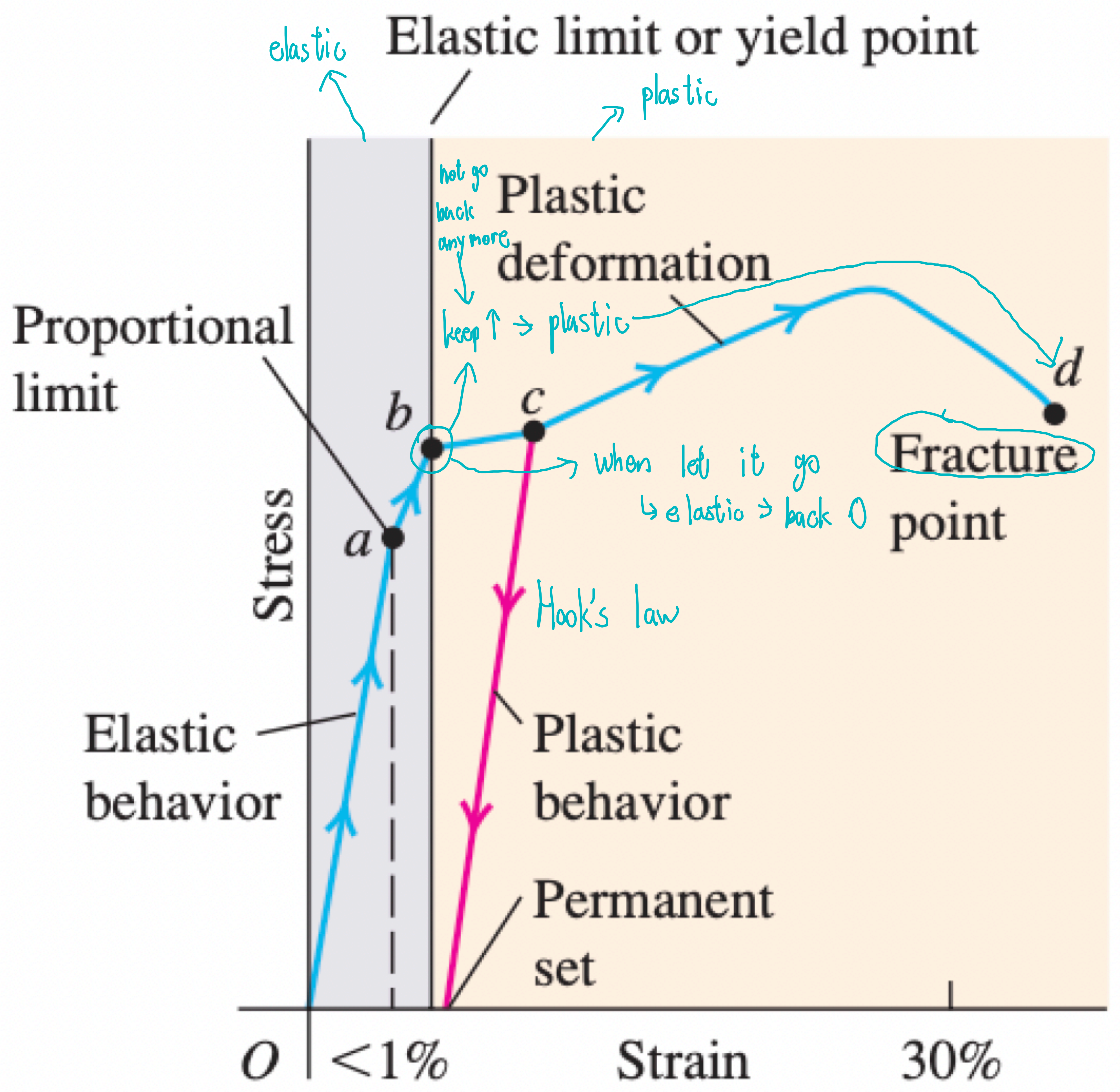
$$\Delta V = \frac{-1.6 \times 10^7 \cdot 0.25}{5.0 \times 10^9}$$

$$= -8 \times 10^{-4} \ m^3$$

Decrease in $8 \times 10^{-4} \ m^3$

Elasticity vs. Placiticity

$stress = modulus \times strain$



shear modulus

$\frac{x}{h} \ll 1$